FW

FLEETWATCH & FLEETSENTINEL

AUTONOMOUS CORRIDOR TELEMETRY, DRIVER COMPLIANCE & RISK GOVERNANCE ENGINE

ENTERPRISE SPECIFICATION

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LEAN TELEMETRY MVP SPECIFICATION & STRATEGIC ENTERPRISE ARCHITECTURE

Autonomous Telemetry, Geofenced Corridor Compliance & Driver Risk Engine

A mission-critical product specification, architectural gap audit, and regulatory defense framework engineered for heavy haulage operations along South Africa's high-risk freight arteries (N4 Highveld Corridor: Pretoria East — Bronkhorstspruit — Witbank).



CORRIDOR SHIELD
300m Buffer
Sub-second N4 mainline & toll skirting detection

EDGE INGRESS
SQLite Cache
Survives 14km Balmoral cellular dead zone

DRIVER SAFETY
RTMS 4.0h
Continuous fatigue governor & in-cab alerts

COMMERCIAL PILOT
R5,000 MVP
4-Week rapid field pilot to retainer upsell

AUTHORED & ARCHITECTED BY
Sifiso Madonsela • Lead Consultant & System Architect
In the words of Principal Product Designer & Head of Product / Product Owner

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Executive Summary & Product Manifesto

The shift from passive dots-on-a-map telematics to active, autonomous operational enforcement.

> *"Traditional telematics sold the illusion of control: a manager staring at a screen of green dots, completely blind to the fact that Truck FW-01 just pulled off the N4 onto an unpaved Balmoral bypass to evade an R182 toll, putting R15M in cargo directly in a syndicate hijack alley while the driver's phone is dying. FleetWatch / FleetSentinel was designed to kill passive monitoring. Our system is an autonomous operational nervous system."*
>
> ***— Sifiso Madonsela, Lead Consultant & System Architect (Product Designer & Owner)***

The South African Heavy Haulage Reality

Commercial road transport in South Africa accounts for over **84% of all freight movement**, carrying high-value coal, chrome, platinum, agricultural exports, and FMCG across thousands of kilometers of national highways. Yet, fleet operators operate on razor-thin operating margins between 4% and 7%, while facing four existential threats along corridors like the N4 Highveld freight artery:

1. Syndicate Fuel Siphoning

R23.50 / LITRE

A 650-litre tank on a Scania R500 represents R15,275 in cash on wheels. Organised siphoning syndicates at unmonitored truck stops drain 50L to 150L in minutes using drop-pipes, costing an average fleet operator over R95,000 per vehicle annually.

2. Toll Evasion & Hijack Corridors

BALMORAL R104 DIRT TRAP

Drivers divert off the N4 onto gravel bypasses like the R104 Balmoral spur to evade SANRAL toll plazas (Diamond Hill @ R182/truck). This unpaved bypass increases transit wear, delays delivery by 40 minutes, and exposes 38-ton coal interlinks to armed syndicates.

3. Driver Fatigue & Statutory Breach

RTMS SANS 1395

Driver exhaustion causes 38% of heavy haulage rollover fatalities. SANS 1395 mandates a strict 4.0-hour continuous driving ceiling. Dispatchers manually calling drivers fail during peak night hours, resulting in AARTO demerit suspensions and insurance repudiations.

4. Unclaimed Tax Incentives

SARS SCHEDULE 6

Customs and Excise Act allows hauliers to claim back R3.65/L for fuel burned during off-road mining depot operations. Without automated engine-hour logbooks cross-referenced to GPS geofences, operators forfeit millions in legitimate tax rebates annually.

Core Product Metrics & Deliverables

STRATEGIC DIMENSION	CURRENT LEGACY BASELINE	FLEETSENTINEL AUTONOMOUS STANDARD	COMMERCIAL IMPACT
Corridor Deviation Detection	15-45 minutes (post-facto audit)	Under 3 seconds (real-time 300m buffer breach)	Elimination of cargo hijack ambush exposure
Liquid Fuel Model / Siphon	Undetected until next depot dip-stick	Immediate trigger on 15L sudden drop or gradient skew	R4.9M saved annually across 50-truck fleet
Driver Rest Compliance	Unverified manual paper logbooks	Automated RTMS continuous drive timer (4.0h limit)	Zero statutory SANS 1395 audit non-conformances
Speed Governance	Generic 120km/h consumer alerts	Class 3 Heavy Vehicle 80 km/h strict lock	60% reduction in AARTO demerit penalty liability
Highveld Signal Blackspots	Telemetry permanently lost in dead zones	Handset SQLite store & forward (batches of 20)	100% audit integrity through 14km Balmoral gap
Desktop Exception Dispatch	Constant dashboard polling fatigue	ntfy.sh SSE push relay with audio chime to desktop	Instant dispatcher response without screen fatigue

Chapter 1: The Product Anatomy — What It Is

A high-precision telemetry, spatial boundary enforcement, and operational exception engine.

> *"From an interaction design perspective, FleetSentinel is an operational autopilot. It does not demand that a controller babysit fifty vehicles. Instead, it codifies spatial geometry, South African road traffic legislation, and physics into an autonomous state machine that surfaces exceptions in sub-second latency."*
>
> ***— Lead Product Designer Perspective***

The Five Foundational Pillars

1. Geofenced Corridor Polylines

Predefined freight corridors (N4 Highveld Artery from Pretoria East through Diamond Hill and Bronkhorstspruit to Witbank Depot) buffered to a strict 300m operational envelope, with authorized off-ramp service polygons (Rayton R515 Petroport).

2. Spatial Verification & Rules Engine

Sub-second point-to-line projection (PostGIS `ST_DWithin` / Turf.js) calculating lateral deviation, speed locks, and corridor transit validity against authorized polygon boundaries.

3. Resilient Handset Edge Ingress

Android Foreground Service polling GPS every 30s while moving (>5 km/h) and 120s when stationary. BatteryManager telemetry + local SQLite cache bridging the 14km Balmoral cellular dead zone.

4. Desktop Push Exception Dispatch

Native desktop push relay powered by ntfy.sh Server-Sent Events (SSE). Bypasses continuous screen monitoring by firing instant audio chimes and system toasts on critical rule breaches.

End-to-End System Architectural Data Flow

```
[EDGE] Android APK / Handset Ingress (Battery % + Charging + Lat/Lng + Speed)
| → Offline SQLite (pings_cache) if in 14km Balmoral cellular dead zone (drains batches of 20)
[INGRESS] FastAPI REST API Gateway (Bearer Token Auth • /api/v1/telemetry/ping)
| → Hot State Table (fleet_units) + Append-Only Exception Ledger (alerts_log)
[EVALUATOR] Automated Rules Engine (Corridor Buffer • RTMS 4h • Phone Cutoff • Idle)
├─► ntfy.sh SSE Relay → Desktop Toast Alerts & Audio Chimes (Dispatchers/CEO)
└─► WebSockets / Realtime → MapLibre 3D Cockpit (Liquid Fuel Wave & Audit Inspector)
```



IN-CAB EDGE EXPERIENCE

Persistent Foreground Service

Ongoing notification banner: *"FleetSentinel Active — N4 Corridor Monitored"*. Whitelisted against OEM task-killers (Samsung, Xiaomi, Transsion), capturing device charging state and GPS health.

Chapter 2: Complete Functional Inventory — What It Can Do

Detailed verification of the operational modules, algorithms, and compliance rules active in the system.

The Four Core Automated Compliance Rules

Rule 1: Corridor & Toll Skirting

>300M BREACH

Point-to-line lateral distance check (`ST_DWithin` / `turf.pointToLineDistance`). If lateral distance exceeds 300m off the N4 polyline without entering an authorized off-ramp polygon (Diamond Hill Toll, Rayton R515 Petroport), triggers `CORRIDOR_BREACH` .

Rule 2: Fatigue Rest Governor

RTMS 4.0H / 240M

Continuous driving duration counter. If consecutive pings with speed >15 km/h accumulate >240 minutes (4.0 hours) without a registered 15-minute rest interval, triggers `FATIGUE_LIMIT_EXCEEDED` to guarantee SANS 1395 compliance.

Rule 3: Transit Phone Cutoff

>120S SIGNAL GAP

In-transit phone tamper monitor. If speed >0 km/h and no ping is received for >120 seconds, triggers `PHONE_OFFLINE_IN_TRANSIT` , instantly surfacing handset power-off, flight mode, or physical dumping.

Rule 4: Excessive Idling

>30M OUTSIDE WAYPOINT

Thermal and stationary monitor. If coordinates remain static (Δ <10m) with speed 0 km/h for >30 minutes outside a registered depot or Petroport waypoint, triggers `EXCESSIVE_IDLE` for diesel waste and covert offloading risk.

Operational Verification & Rules Matrix

RULE IDENTIFIER	MATHEMATICAL / LOGIC CONDITION	SEVERITY	SYSTEM ACTION & DISPATCHER RELAY	RISK / COST EXPOSURE
CORRIDOR_BREACH	Lateral offset > 300m off N4 polyline AND NOT in authorized polygon	CRITICAL	ntfy.sh desktop push with audio chime + Red pin on 3D Cockpit	R15,000,000 cargo hijack risk
TOLL_SKIRTING	Entering Balmoral R104 dirt trap to evade Diamond Hill toll	HIGH	Instant alert to dispatch + driver WhatsApp warning	R182 toll evasion + R2,500 wear
FATIGUE_LIMIT_EXCEEDED	Rolling continuous driving duration > 240m @ speed > 15 km/h	HIGH	RTMS non-conformance logged + mandatory rest directive sent	SANS 1395 statutory liability
PHONE_OFFLINE_IN_TRANSIT	Last ping age > 120s while last speed > 0 km/h	CRITICAL	ntfy.sh audio chime: "Phone offline in transit — possible tampering"	Covert device disabling risk
EXCESSIVE_IDLE	Static position (Δ<10m) & speed=0 for > 30m outside waypoint	MEDIUM	Diesel burn warning (3.5L/hr wasted) + driver inquiry	R82.25/hr diesel burn waste
SPEED_LIMIT_EXCEEDED	Speed > 80 km/h on national road for > 15 seconds	MEDIUM	In-cab warning audio ping + AARTO penalty counter updated	AARTO demerit points & tyre wear

Cockpit Visual Intelligence Modules

3D Orthophoto MapLibre

WebGL canvas with 45-degree pitch, high-res satellite tiles, N4 corridor buffer geometry, and dynamic Haversine 1.22 road tortuosity calculations.

Liquid Fuel Model (34L/100km)

CSS wave animation driven by real-time distance burn math, instantly flagging irregular fuel drops against expected consumption curves.

Driver Audit Inspector

Four live status badges reflecting Toll Status, Continuous Drive Hours, Handset Battery & Charging State, and Telemetry Heartbeat.

Chapter 3: What Is Missing — The Brutal Gap Audit

A candid assessment of current technical debt, missing enterprise capabilities, and scaling bottlenecks.

> *"As Product Owner, I refuse to sugarcoat technical reality. What we have built today in FleetWatch / FleetSentinel is a brilliant operational architecture, but it currently relies on simulated telemetry and single-process Node memory. To scale to a commercially defensible enterprise product, we must brutally confront the gaps across the edge, backend, and hardware layers."*
>
> ***— Product Owner & Lead Architect Reality Check***

The Four Critical Architectural Gaps

1. Missing Native Mobile APK & Offline Caching

Currently, driver pings are simulated in seed data. In reality, between Bronkhorstspruit and Balmoral, cellular coverage drops for 14 km. Without an Android Foreground Service and local SQLite buffer draining in batches of 20, signals are lost, triggering false offline panics.

2. Ephemeral In-Memory Store vs PostGIS

The current server stores telemetry in JavaScript memory (`store.js`). A container restart wipes historical trip tracks and alerts. Production requires PostgreSQL + PostGIS with persistent `fleet_units` and `alerts_log` tables.

3. Pure Software Math vs Physical Probes

Fuel levels are currently estimated via distance ($34 ext{L}/100 ext{km}$). An actual syndicate siphon cannot be proven in court without physical capacitive/acoustic RS485 fuel level probes and CAN-bus J1939 injector flow telemetry.

4. Missing Multi-Tenant RBAC & Workflows

No driver vs dispatcher authentication roles. Missing automated PDF trip manifest exports, driver incident dispute ticketing, and digital electronic proof-of-delivery (ePOD) sign-offs.

Detailed Gap & Remediation Priority Matrix

SUBSYSTEM LAYER	CURRENT STATE (PROTOTYPE / MOCK)	REQUIRED PRODUCTION STATE	IMPACT & URGENCY
Edge Ingress (Mobile)	Simulated Node script (journey_mule.js)	Android APK Foreground Service + SQLite cache for 14km Balmoral dead zone	URGENT: Core prerequisite for live field pilot
Ingress API & DB	Single Node Express with RAM store	FastAPI + Supabase / PostgreSQL + PostGIS spatial buffering	HIGH: Multi-truck scalability & crash persistence
Push Dispatcher	Local browser console / WebSocket only	ntfy.sh Server-Sent Events (SSE) desktop push relay with sound	HIGH: Eliminates dispatcher screen monitoring fatigue
Fuel Telemetry	Theoretical mathematical burn model	Dual RS485 capacitive fuel tank probes with Kalman slosh filter	MEDIUM: Enterprise upsell tier beyond Lean MVP
Engine Diagnostic	Calculated speed and GPS estimates	Contactless J1939 inductive CAN-bus clamp (FMS standard)	MEDIUM: Enterprise tier; shelved for MVP launch
Vehicle Security	Passive visual alert on UI canvas	Remote starter cut-off relay with dual-authorization speed interlock	MEDIUM: Future security tier; safety-critical

Chapter 4: Enterprise Cyber Defense & Telematics Security

Zero-trust device authentication, GPS anti-jamming heuristics, and fail-safe vehicle immobilization controls.

Threat Modeling: South African Syndicate Attack Vectors

In South Africa, telematics systems face sophisticated physical and electronic attack vectors from well-funded syndicates operating along the N4 and N3 corridors. A viable security architecture must defend against three primary threat categories:

1. GPS Jamming & Spoofing

12V cigarette-lighter multi-band GPS jammers flooding L1/L2 frequencies, creating false stationary readings or complete signal blackout during syndicate highjack ambushes.

2. Handset Tampering & Mocking

Drivers using Android Mock Location apps, toggling flight mode, or wrapping phones in foil bags to covertly conduct unauthorized side-trips and diesel siphoning.

3. Man-in-the-Middle Ingress

Intercepting unauthenticated HTTP ping endpoints, forging artificial speed and coordinate logs to falsify RTMS driving hours and SARS tax rebate submissions.

Enterprise Security Defense Specification

SECURITY VECTOR	VULNERABILITY EXPOSURE	FLEETSENTINEL DEFENSIVE ARCHITECTURE	VERIFICATION STANDARD
Device Ingress Auth	Forged HTTP telemetry pings	Bearer Token / HMAC-SHA256 device key bound to Android Keystore	Zero unauthenticated packet acceptance
Anti-Spoofing & Mocking	Android Mock Location provider apps	Play Integrity API + GPS vs Cell-Tower Doppler jitter validation	Auto-reject pings flagged by Mock Provider
Tamper-Evident Logging	Dispatcher altering logs to hide theft	Chained SHA-256 append-only ledger in alerts_log; immutable audit trail	Court-admissible cryptographic proof
Data In-Transit & At-Rest	Sniffing cellular pings or DB breach	TLS 1.3 with certificate pinning; AES-256 encryption on DB & SQLite	POPIA Section 19 compliance certified
Vehicle Immobilization	Syndicate hijacking truck in transit	Dual-authorization remote starter cut-off (interlocked to speed < 5 km/h)	Strict fail-safe (never cut engine at speed)
Role-Based Access (RBAC)	Privilege escalation by rogue staff	Principle of Least Privilege: Dispatcher, Safety Officer, Executive, Driver	MFA required for immobilization & geofencing

Remote Immobilization Safety Architecture

SAFETY-CRITICAL PNEUMATIC & STARTER INTERLOCK — FAIL-SAFE PROTOCOL

Remotely killing a 56-ton Super-Link moving at 80 km/h will cause catastrophic loss of pneumatic steering and air-brake pressure, resulting in fatal rollover collisions. FleetSentinel's immobilization protocol enforces a **Three-Tier Hardware Interlock**:

```
1. Verified Speed Check: Speed must be confirmed at 0 km/h (or < 5 km/h) across two consecutive pings.
2. Dual Supervisor Authentication: Digital biometric or TOTP authorization required from Operations Director.
3. Starter Relay Cut (Non-Engine Cut): Isolates starter solenoid, allowing engine to idle for cooling but preventing restart once switched off.
```

Chapter 5: Regulatory Compliance & Legal Framework

Aligning telemetry algorithms with South African statutory legislation, POPIA privacy, and RTMS standards.

1. POPIA (Protection of Personal Information Act 4 of 2013)

Under South African law, continuous telemetry tracking, driver handset battery monitoring, and behavioral scoring constitute **Special Personal Information (SPI)**. Failure to comply exposes operators to administrative fines up to **R10 Million** and imprisonment under Section 107. FleetSentinel guarantees compliance through the 8 Lawful Processing Conditions:

Condition 2: Processing Limitation

Explicit operator-driver contractual consent clauses; monitoring strictly restricted to active shift hours. Automatic suspension of location tracking when driver logs out off-shift.

Condition 3: Purpose Specification

Data collected solely for cargo safety, RTMS driver fatigue compliance, and statutory tax rebates. Purpose documented in the enterprise Operator Privacy Policy.

Condition 7: Security Safeguards

AES-256 database encryption, role-based access control, cryptographic key rotation, and automated Section 22 data breach notification mechanisms.

Statutory Information Officer

Mandatory registration of the Fleet Information Officer with the Information Regulator, establishing formal dispute, rectification, and subject access procedures.

2. RTMS (Road Transport Management System — SABS ARP 067 / SANS 1395)

RTMS is the voluntary national heavy haulage accreditation standard recognized by SANRAL, mining houses, and major insurers. FleetSentinel automates the three critical compliance pillars:

RTMS STANDARD DIMENSION	SANS 1395 REGULATORY MANDATE	FLEETSENTINEL AUTONOMOUS ALGORITHM	AUDIT PROOF DELIVERABLE
Driver Hours & Fatigue	Max 4.0h continuous driving; min 15m rest break; max 14h shift	Rule 2: Rolling timer tracking speed >15km/h; flags breach at 240m	Automated RTMS Driver Fatigue Logbook with rest verification
Speed Governance	Strict adherence to 80 km/h heavy freight speed limits	Real-time GPS overspeed counter; triggers at >80 km/h for >15s	Speed Exception Digest & Driver Demerit Scorecard
Vehicle Roadworthiness	Mandatory daily pre-trip walkaround inspection	Mobile APK digital inspection checklist before ignition unlock	Timestamped digital vehicle safety cert with photo upload
Overload & Axle Mass	Compliance with Bridge Formula & Legal Axle Limits	Integration with bridge scale weight tickets via API	Overload prevention log defending against impoundment

Commercial Value of RTMS Accreditation INSURANCE & TOLL CONCESSIONS

Accredited fleets receive up to **15% reductions on commercial goods-in-transit insurance premiums**, preferential loading access at major Anglo American / Glencore coal terminals, and bypass privileges at SANRAL weighbridges.

Chapter 5 (Cont.): Hazchem, AARTO, Tax & SADC

Expanding statutory defenses across dangerous goods, traffic demerit points, and customs rebate logbooks.

1. SARS Diesel Refund Scheme (Customs & Excise Act 91 of 1964, Schedule 6)

Mining and agricultural hauliers are entitled to refund claims of up to **R3.65 per litre** for fuel consumed in non-public road operations. However, SARS aggressively audits and disallows claims lacking rigorous logbooks. FleetSentinel generates **Audit-Proof Digital Logbooks**:

Geofenced Mining Pit Demarcation

Automatically detects when a coal tipper crosses into an authorized off-road pit/quarry geofence. Clocks off-road engine hours and calculates qualifying diesel burn.

Automated Schedule 6 Logbook Export

One-click generation of SARS-compliant XML/PDF schedules cross-referencing fuel receipts, GPS entry/exit timestamps, and engine hours, defending millions in rebates.

2. NRTA & AARTO (Demerit Points Defense)

Under the Administrative Adjudication of Road Traffic Offences (AARTO) Act, heavy vehicle operators face license suspensions if drivers accumulate demerit points. FleetSentinel acts as the corporate legal shield:

LEGISLATION	STATUTORY REQUIREMENT	FLEETSENTINEL TECHNICAL ENFORCEMENT	OPERATIONAL BENEFIT
NRTA 93 of 1996 Reg 292	Goods vehicle >9,000kg restricted to 80 km/h on freeways	Real-time speed lock algorithm alerts dispatch if speed exceeds 80 km/h	Prevents statutory reckless driving citations
NRTA Reg 115	Driver must possess valid PrDP (Category G or D)	Driver profile database blocks dispatch if PrDP is expired or unverified	Eliminates unlicenced driver insurance repudiation
AARTO Demerit System	Demerit points attributed to operator for driver infringements	Timestamped GPS corridor audit logs provide forensic defense for false fines	Defends operator fleet operating licence from suspension
SANS 10231 Hazchem	Mandatory Tremcards, placarding & designated corridor adherence	Corridor geofences alert if dangerous goods tankers divert into residential zones	Full compliance with Disaster Management protocols
SADC Cross-Border Protocol	Cross-border cargo tracking (Lebombo / Maputo Corridor)	Multi-country SIM roaming & customs transit seal geofences at border posts	Seamless transit between South Africa & Mozambique

Cross-Border & SADC Corridor Protocol

For hauliers moving chrome and coal along the N4 Maputo Corridor across the Lebombo/Ressano Garcia border post into Mozambique, FleetSentinel incorporates **Customs Transit Geofencing**. It monitors container door seals, tracks clearing agent dwell times, and switches seamlessly across Vodacom SA and Vodacom Mozambique networks without losing a single telemetry ping.

Chapter 6: Telematics Hardware, CAN-Bus & Edge Protocols

Bridging low-cost mobile handset edge ingress with industrial telematics gateway hardware.

The Dual-Tier Edge Hardware Architecture

To balance immediate commercial viability with future enterprise scalability, FleetSentinel defines a modular two-tier edge hardware strategy:

Tier 1: Handset Edge Ingress (Lean MVP Launch) ZERO HARDWARE CAPEX

Utilizes driver Android handsets with persistent Foreground Service, BatteryManager telemetry, and local SQLite caching. Delivers 90% of corridor and driver compliance value at zero vehicle installation cost.

Tier 2: Industrial CAN-Bus Telematics (Enterprise Upsell) INDUSTRIAL IOT

Dedicated Teltonika FMB640 / Ruptela gateways with contactless inductive CAN clamps (J1939 FMS), RS485 capacitive fuel probes, and dual starter immobilization relays.

Telematics Hardware Integration Specifications

COMPONENT / SENSOR	INDUSTRIAL PROTOCOL	TELEMETRY DATA HARVESTED	OPERATIONAL ROLE
Driver Android Handset	HTTPS / WSS JSON REST	GPS Lat/Lng, Speed, Battery %, Charging State, Handset Heartbeat	Tier 1 Lean MVP ingress; zero installation cost
Contactless CAN Clamp	J1939 FMS Inductive (ISO 11898)	Engine RPM, True Odometer, Fuel Consumed, Engine Torque %	Reads powertrain data without splicing cables or voiding OEM warranty
Capacitive Fuel Probes	RS485 Modbus RTU (Dual-Tank)	Millimeter-accurate liquid diesel level with temperature compensation	Detects stationary drop-pipe siphoning with 99.2% accuracy
Starter Isolation Relay	12V / 24V Automotive Solenoid Relay	Closed-loop starter circuit kill interlocked to speed < 5 km/h	Safe anti-theft immobilization; prevents vehicle restart
Industrial Gateway (FMB640)	MQTT / TCP Binary Protocol	Buffered flash storage (16MB), dual Micro-SIM, internal backup battery	Hardware backbone surviving complete vehicle battery disconnect
Satellite Hybrid Modem	Iridium SBD / Starlink Direct-to-Cell	Low-frequency burst pings (1 ping / 15m) when out of cellular range	Cross-border backup for remote African mineral transit corridors

Contactless CAN-Bus Inductive Clamping

Traditional telematics installers cut into heavy vehicle wiring looms to tap CAN-High and CAN-Low wires, frequently voiding **Scania, Volvo, and Mercedes-Benz factory warranties**. FleetSentinel mandates **Contactless Inductive Magnetic Clamps** that read digital CAN pulses through wire insulation via magnetic induction, guaranteeing zero electrical footprint and zero warranty interference.

Chapter 7: Product Experience & Design System

Designing for high-stress operational cockpits: cognitive ergonomics, visual hierarchy, and spatial clarity.

> *"In a fleet control room at 02:00 AM, a dispatcher is managing dozens of vehicles on a dimly lit multi-monitor desk. If your interface looks like an Excel spreadsheet covered in tiny red dots, the dispatcher will miss critical events due to alarm fatigue. Every pixel in FleetSentinel is engineered to minimize cognitive friction and amplify immediate situational awareness."*
>
> ***— Lead Product Designer UX Manifesto***

The Three Ergonomic Design Principles

1. 3-Tier Alert Color Hierarchy

Emerald Green (`#10b981`) for full compliance. Amber (`#f59e0b`) for advisory alerts. Flashing Crimson (`#ef4444`) reserved exclusively for critical corridor breaches and siphoning.

2. Spatial GIS Affordance

MapLibre 3D tilted orthophoto canvas with illuminated 300m corridor bounds. Trucks are represented with bearing-oriented directional icons and smooth coordinate interpolation.

3. Liquid Physical Metaphor

Dynamic CSS liquid wave gauge providing an intuitive physical representation of remaining tank capacity, burning realistically with traveled corridor distance.

Color Tokens & State Semantics

DESIGN TOKEN	HEX VALUE	VISUAL WEIGHT	OPERATIONAL MEANING & APPLICATION
Status Compliant	#10b981	Vibrant Emerald	Truck strictly inside 300m corridor, RTMS rest compliant, normal fuel gradient.
Status Advisory	#f59e0b	Vivid Amber	Dwell time > 20 mins, continuous drive time approaching 3.5h, speed 75–80 km/h.
Status Critical	#ef4444	Crimson Red	Corridor breach (>300m), Balmoral toll diversion, phone offline in transit, siphon.
Corridor Buffer Line	#38bdf8	Cyan Polyline	Illuminated 300m N4 mainline operational envelope with dashed tolerance zone.
Canvas Background	#0b1120	Deep Slate	High-contrast dark mode reducing eye fatigue during 12-hour night shifts.

Driver Audit Inspector Status Badges

BADGE 1
Toll Status
Diamond Hill: Cleared

BADGE 2
Drive Timer
2.4h / 4.0h RTMS

BADGE 3
Battery & State
88% • Charging

BADGE 4
Heartbeat
Ping: 2s ago

Chapter 8: Target Enterprise Architecture & Data Engineering

A horizontally scalable, distributed event-driven blueprint capable of handling 50,000+ active freight units.

Target Event-Driven Microservices Architecture



Database Schema Specification (PostgreSQL + PostGIS)

Table 1: fleet_units (Hot State Table)

Maintains the latest validated telemetry point per registered truck.

```
truck_reg VARCHAR(20) PRIMARY KEY
driver_id UUID REFERENCES drivers(id)
current_geom GEOMETRY(Point, 4326)
speed_kmh NUMERIC(5,2)
battery_pct SMALLINT
is_charging BOOLEAN
continuous_drive_minutes INT
last_ping_timestamp TIMESTAMPTZ
```

Table 2: alerts_log (Append-Only Ledger)

Immutable, audit-proof record of every rule violation and resolution.

```
alert_id UUID PRIMARY KEY DEFAULT gen_random_uuid()
truck_reg VARCHAR(20) NOT NULL
rule_type VARCHAR(50) NOT NULL
severity VARCHAR(20) NOT NULL
lateral_offset_meters NUMERIC(6,1)
trigger_geom GEOMETRY(Point, 4326)
created_at TIMESTAMPTZ DEFAULT NOW()
sha256_hash VARCHAR(64) -- Audit proof
```

High-Availability & Disaster Recovery SLA

RELIABILITY DIMENSION	TARGET SLA	ARCHITECTURAL IMPLEMENTATION
System Availability	99.95% Annual Uptime	Multi-AZ cloud deployment on AWS Cape Town (af-south-1) with auto-scaling
Recovery Point Objective (RPO)	< 1.0 Minute	Continuous WAL archiving to redundant S3 buckets with cross-region sync
Recovery Time Objective (RTO)	< 15.0 Minutes	Automated Kubernetes pod failover and standby read-replica promotion
Offline Edge Buffer Longevity	72 Hours of Telemetry	SQLite on-device ring buffer holding up to 8,640 compressed pings per vehicle

Chapter 9: Phased Commercial & Technical Roadmap

A structured 4-week Lean Telemetry MVP pilot schedule evolving into a 12-month enterprise platform.

Part A: Lean Telemetry MVP — 4-Week Sprint Schedule (FleetSentinel Launch)

SPRINT & TIMELINE	CORE OBJECTIVES	TECHNICAL TASKS & DELIVERABLES	COMMERCIAL MILESTONE
Week 1: Core Engine (Days 1–7)	Cloud Backend & PostGIS Rules Engine	• Deploy FastAPI Ingestion API (/api/v1/telemetry/ping) • Provision PostgreSQL tables (fleet_units, alerts_log) • Program the 4 rules: 300m Corridor, RTMS 4h, Cutoff, Idle • Configure ntfy.sh desktop push relay with priority headers	Internal pipeline tests on simulated N4 journey data
Week 2: Mobile APK (Days 8–14)	Driver Android APK & Edge Buffer	• Persistent Android Foreground Service with ongoing notification • OEM battery killer bypass (Samsung, Xiaomi, Transsion) • GPS polling: 30s moving / 120s static + BatteryManager % • Handset SQLite buffer draining batches of 20 in 14km Balmoral gap	Successful bench testing of offline queue synchronization
Week 3: Integration (Days 15–21)	Cockpit Wiring & Real-Time Visualization	• Connect MapLibre 3D WebGL canvas to live WebSocket stream • Dynamic Haversine 1.22 tortuosity & liquid fuel burn model • Desktop ntfy.sh SSE toast receiver with audio chimes • Driver Audit Inspector panel with 4 live status badges	Complete end-to-end integration walkthrough for client
Week 4: Field Pilot (Days 22–28)	Live N4 Road Test & Client Deployment	• Live field test on N4 corridor (Pretoria East to Balmoral) • Handset deployment in test truck cab; test Balmoral detour • Review exception alerts with haulier Managing Director	R5,000 Baseline Pilot Signed ➔ Retainer Upsell

Part B: Scope Boundaries (What We Build vs What We Shelve for Launch)

IN-SCOPE: LEAN TELEMETRY MVP LAUNCH

- Driver Android APK with Foreground Service & BatteryManager
- Handset SQLite store & forward queue (14km Balmoral gap)
- FastAPI + PostgreSQL PostGIS 300m corridor buffering
- The 4 Automated Rules (Corridor, RTMS 4h, Phone Cutoff, Idle)
- Live 3D MapLibre Cockpit & ntfy.sh Desktop Push Relay

SHELVED: FUTURE ENTERPRISE TIERS (NOT FOR LAUNCH)

- Inductive CAN-bus J1939 powertrain clamps (FMS standard)
- Physical digital capacitive/acoustic fuel tank probes
- Remote starter cut-offs & dual-authorization relays
- Complex satellite hybrid modems (Iridium SBD)
- Multi-country cross-border clearing integrations

Part C: 12-Month Enterprise Evolution Horizon

HORIZON PHASE	KEY DELIVERABLES & TECHNOLOGY	TARGET COMMERCIAL OUTCOME
Phase 1: Enterprise Hardening (Mo 1–3)	PostgreSQL 16 + PostGIS persistence; Kafka ingestion broker; mTLS device keys; immutable SHA-256 audit ledger.	Production readiness for 250-truck fleet rollouts.
Phase 2: Hardware Telematics (Mo 4–6)	J1939 CAN-bus inductive clamp ingestion; Teltonika FMB640 gateway support; RS485 fuel probes.	Elimination of driver phone dependency.
Phase 3: Regulatory Suites (Mo 7–9)	RTMS SANS 1395 digital auditor pack; pre-trip walkaround checklist PWA; AARTO demerit tracking.	Accreditation as an Approved RTMS Provider.
Phase 4: SADC Cross-Border (Mo 10–12)	C-BRTA Maputo/Beitbridge tracking; multi-currency fuel & toll ledger; predictive AI hijack risk heatmaps.	Enterprise licensing to top-tier JSE-listed miners.

Chapter 10: Financial Impact, Unit Economics & ROI Model

Demonstrating tangible cash flow preservation and risk reduction for a representative 50-truck heavy haulage fleet.

> *"As Head of Product, my job is not just to deliver beautiful software; it is to hand our clients' CFOs a mathematically unassailable business case. A 50-truck fleet operating on the N4 Highveld Corridor spends approximately R45,000,000 annually on diesel and maintenance. FleetSentinel generates a 680% return on investment with a cash payback period under 42 days."*
>
> ***— Head of Product / Commercial Strategist***

Annual Cost-Benefit Model: 50-Truck Heavy Bulk Fleet (N4 Corridor)

FINANCIAL VALUE DRIVER	BASELINE ANNUAL LOSS (UNPROTECTED)	FLEETSENTINEL PROTECTED PERFORMANCE	NET ANNUAL SAVINGS (ZAR)
Fuel Theft & Siphoning Prevention	350L siphoned / truck / mo @ R23.50/L = R4,935,000 / yr	94% theft reduction via liquid model & corridor alert	R4,638,900
SANRAL Toll Evasion Avoidance	15 bypasses/mo @ R2,500 damage & fine = R450,000 / yr	Zero unauthorized Balmoral diversions; 100% N4 adherence	R450,000
SARS Schedule 6 Diesel Rebate	Unclaimed off-road mining hours due to lack of proof	Automated engine-hour logbooks claiming R3.65/L rebate	R1,150,000
Insurance Premium Concession	Standard heavy haulage fleet premium @ R3,200,000 / yr	15% underwriter discount for RTMS & active corridor shield	R480,000
Tyre & Powertrain Wear (80 km/h)	Excessive wear driving at 95-105 km/h on Donkerhoek pass	Enforced 80 km/h heavy freight lock extending tyre life 22%	R320,000
AARTO Demerit Fine Elimination	Average R15,000/month in traffic fines across 50 drivers	Real-time in-cab speed buzzer reducing speeding by 85%	R153,000
TOTAL ANNUAL VALUE CREATED	Combined Annual Losses: R9,215,000	Total Protected Value Realization	R7,191,900

Fleet Investment & TCO Breakdown

Commercial Deployment Model

R5,000 Baseline Pilot: 4-week field validation on test trucks along N4.
Production Retainer Upsell: R650 / truck / month SaaS subscription.
Cellular Data & Server Infra: R120 / truck / month SIM & cloud relay.
Optional Tier 2 Hardware CapEx: R12,500 / truck (probes & CAN clamps).

Year 1 Total Investment (SaaS + Pilot + Infra): R467,000



680% ROI

NET FIRST-YEAR RETURN ON INVESTMENT • PAYBACK: 42 DAYS

Appendix: Technical Reference & Codebase Catalog

Exhaustive REST API endpoints, WebSocket event contracts, coordinate systems, and domain glossary.

Core REST API Endpoints Catalog

METHOD & ENDPOINT	PAYLOAD / QUERY PARAMETERS	RESPONSE CONTRACT	SECURITY & DESCRIPTION
POST /api/v1/telemetry/ping	{driver_id, truck_reg, lat, lng, speed_kmh, battery_pct, is_charging, timestamp}	{status: "ok", corridor_status, zone, violations: []}	Bearer token auth; validates 300m buffer, fatigue & idle
POST /api/v1/telemetry/batch	{driver_id, truck_reg, pings: [...]} (array of up to 20 pings)	{status: "ok", processed: 20, buffered_violations: []}	Drains offline SQLite cache after Balmoral cellular dead zone
GET /api/v1/fleet/trucks	None (or ?status=active)	{success: true, count: N, trucks: [...]}	Returns hot-state fleet units with live coordinates & badges
POST /api/v1/fleet/corridors/exceptions	{name, offRampPoint, onRampPoint, polygon, maxDwell}	{success: true, exception_id: UUID}	Whitelists authorized service bypasses (e.g. Rayton Petroport)
GET /api/v1/fleet/reports/sars-diesel-logbook	?taxYear=2026&truck_reg=FW-01	{taxYear, totalRebateZAR, logs: [...]}	Generates official Schedule 6 diesel refund claims
POST /api/v1/fleet/simulate/:type	:type = "siphon" | "overspeed" | "corridor-detour"	{success: true, simulatedType, alert}	Executive demo injector for field pilot walkthroughs

WebSocket & SSE Event Bus

```
// Real-Time Emitters (Socket.io & ntfy.sh SSE)
telemetry_ping → Live position & speed interpolation
corridor_breach → Lateral offset > 300m off N4
fatigue_warning → RTMS 4.0h limit exceeded
phone_cutoff → Gap > 120s with speed > 0
excessive_idle → Static > 30m outside waypoint
```

Geodetic Coordinate Standards

Global Spatial Standard: WGS84 (EPSG:4326) utilized for MapLibre tiles, GPS ingress, and OSRM routing.
Cadastral Cadence: Hartebeesthoek94 (Lo29 Central Meridian) used for millimeter-accurate coal pit depot boundary alignment in Mpumalanga.

Domain Glossary & South African Logistics Terminology

RTMS: Road Transport Management System (SANS 1395) heavy haulage safety accreditation. • **AARTO:** Administrative Adjudication of Road Traffic Offences Act. • **PrDP:** Professional Driving Permit (Category G / D). • **Super-Link:** Tri-axle tractor pulling twin interlink trailers (56t GVM). • **Tremcard:** Transport Emergency Card mandated under NRTA Chapter VIII for hazardous materials. • **Balmoral Bypass:** Unpaved gravel detour (R104) used by errant drivers to evade Diamond Hill toll plaza.